

Name: _____
Date: _____ Per: _____

Algebra 2
Unit 7: Simplifying Polynomials

Quiz 7-1: Polynomial Operations & Radicals

Multiple Choice Version

Circle the letter of the best answer for each question.

For questions 1–4, use the polynomial: $-2x^2 + x^4 + 3$

1. What is the polynomial written in standard form?

A) $x^4 - 2x^2 + 3$

C) $-2x^2 + x^4 + 3$

B) $3 + x^4 - 2x^2$

D) $3 - 2x^2 + x^4$

2. What is the degree of the polynomial?

A) 2

B) 3

C) 4

D) 5

3. How is the polynomial classified by number of terms?

A) Monomial

B) Binomial

C) Trinomial

D) Polynomial

4. How is the polynomial classified by degree?

A) Quadratic

B) Cubic

C) Quartic

D) Quintic

For questions 5–7, use the polynomial: $-7 + 5m^3 + 9m - 3m^2$

5. What is the polynomial written in standard form?

A) $-7 + 9m - 3m^2 + 5m^3$

C) $5m^3 + 9m - 3m^2 - 7$

B) $5m^3 - 3m^2 + 9m - 7$

D) $-3m^2 + 5m^3 + 9m - 7$

6. What is the degree and leading coefficient of the polynomial?

A) Degree 3, leading coefficient 5

C) Degree 2, leading coefficient -3

B) Degree 3, leading coefficient -3

D) Degree 4, leading coefficient 5

7. How is the polynomial classified by number of terms?

A) Monomial

B) Binomial

C) Trinomial

D) Polynomial (4+ terms)

Simplify. All answers should be in standard form with only positive exponents.

8. Simplify: $(-6a^4b^5)^2 \cdot 2ab^5$

A) $72a^8b^{15}$

C) $72a^9b^{15}$

B) $-72a^9b^{15}$

D) $-72a^8b^{10}$

9. Simplify: $\frac{8x^{-3}y \cdot 5x^5y}{10x^6y^2}$

A) $4x^4$

B) $\frac{4}{x^4}$

C) $\frac{4x^4}{y}$

D) $\frac{4}{x^4y}$

10. Simplify: $(6x^2 + 7x) + (5 - 2x^2 + 9x)$

A) $8x^2 + 16x + 5$

C) $4x^2 + 16x - 5$

B) $4x^2 - 2x + 5$

D) $4x^2 + 16x + 5$

11. Simplify: $(4h^3 + 6h^2 + 1) - (5h^2 - 8h^4 + 6)$

A) $8h^4 + 4h^3 + 11h^2 - 5$

C) $-8h^4 + 4h^3 + 11h^2 + 7$

B) $8h^4 + 4h^3 + h^2 - 5$

D) $8h^4 + 4h^3 + h^2 + 7$

12. Simplify: $(x + 2)(2x^2 - x - 9)$

A) $2x^3 - 3x^2 - 11x - 18$

C) $2x^3 + 3x^2 - 11x - 18$

B) $2x^3 + 3x^2 - 7x - 18$

D) $2x^3 + 3x^2 - 11x + 18$

13. Simplify: $\frac{21c^{10}d^3 + 56c^6d^2 - 7c^2d}{7c^2d}$

A) $3c^{12}d^4 + 8c^8d^3 - 1$

C) $14c^8d^2 + 49c^4d - 1$

B) $3c^5d^2 + 8c^3d - 1$

D) $3c^8d^2 + 8c^4d - 1$

14. A trapezoid has parallel bases of $15x^2y$ and $9x^2y$, with a height of $4x^5$. What is the area of the trapezoid in simplified form?

(Hint: Area of trapezoid = $\frac{1}{2}(b_1 + b_2)h$)

A) $96x^7y$

C) $24x^7y$

B) $48x^7y$

D) $48x^{10}y$

Rewriting and Simplifying Radicals

(TEKS 2A.7B, 2A.7G)

Convert between radical and rational exponent form, then simplify.

15. Which expression is equivalent to $\sqrt{x}$?
- A) x^2 B) $x^{1/2}$ C) x^{-2} D) $2x$
16. Which expression is equivalent to $\sqrt[3]{y}$?
- A) y^3 B) $y^{1/3}$ C) $3y$ D) $\frac{y}{3}$
17. Which expression is equivalent to $x^{2/3}$?
- A) $\sqrt[3]{x^2}$ B) $\sqrt{x^3}$ C) $\frac{2x}{3}$ D) $\frac{2}{3}x$
18. What is the value of $8^{2/3}$?
- A) 2 B) 4 C) $\frac{16}{3}$ D) 12
19. What is the value of $27^{2/3}$?
- A) 3 B) 9 C) 18 D) 729
20. Simplify: $x^{1/2} \cdot x^{1/2}$
- A) $x^{1/4}$ B) x C) x^2 D) $2x^{1/2}$
21. Simplify: $y^{3/4} \cdot y^{1/4}$
- A) $y^{3/16}$ B) y C) $y^{1/2}$ D) $y^{4/4}$
22. Simplify: $(a^{1/4})^8$
- A) a^2 B) a^{32} C) $a^{1/32}$ D) $8a^{1/4}$
23. What is the value of $16^{-3/4}$?
- A) -12 B) -8 C) $\frac{1}{8}$ D) 8
24. Which expression is equivalent to $\sqrt{x} \cdot \sqrt[3]{x}$?
- A) $x^{1/6}$ B) $x^{2/5}$ C) $x^{5/6}$ D) $x^{6/5}$
25. The expression $(27x^6)^{2/3}$ is equivalent to which of the following?

A) $3x^4$

B) $9x^4$

C) $9x^9$

D) $18x^4$

ANSWER KEY – TEACHER COPY

Polynomial Operations (Questions 1–14)

- | | |
|------|-------|
| 1. A | 8. C |
| 2. C | 9. B |
| 3. C | 10. D |
| 4. C | 11. B |
| 5. B | 12. C |
| 6. A | 13. D |
| 7. D | 14. B |

Radicals & Rational Exponents (Questions 15–25)

- | | |
|-------|-------|
| 15. B | 21. B |
| 16. B | 22. A |
| 17. A | 23. C |
| 18. B | 24. C |
| 19. B | 25. B |
| 20. B | |

Detailed Solutions:

- $x^4 - 2x^2 + 3$ (descending order of exponents)
- Degree 4 (highest exponent)
- Trinomial (3 terms)
- Quartic (degree 4)
- $5m^3 - 3m^2 + 9m - 7$ (descending order)
- Degree 3, leading coefficient 5
- 4 terms = polynomial
- $(-6a^4b^5)^2 \cdot 2ab^5 = 36a^8b^{10} \cdot 2ab^5 = 72a^9b^{15}$
- $\frac{8x^{-3}y \cdot 5x^5y}{10x^6y^2} = \frac{40x^2y^2}{10x^6y^2} = \frac{4}{x^4}$

10. $(6x^2 + 7x) + (5 - 2x^2 + 9x) = 4x^2 + 16x + 5$
11. $(4h^3 + 6h^2 + 1) - (5h^2 - 8h^4 + 6) = 8h^4 + 4h^3 + h^2 - 5$
12. $(x + 2)(2x^2 - x - 9) = 2x^3 - x^2 - 9x + 4x^2 - 2x - 18 = 2x^3 + 3x^2 - 11x - 18$
13. $\frac{21c^{10}d^3 + 56c^6d^2 - 7c^2d}{7c^2d} = 3c^8d^2 + 8c^4d - 1$
14. Area = $\frac{1}{2}(15x^2y + 9x^2y)(4x^5) = \frac{1}{2}(24x^2y)(4x^5) = 48x^7y$
15. $\sqrt{x} = x^{1/2}$ — The square root equals exponent of $\frac{1}{2}$
16. $\sqrt[3]{y} = y^{1/3}$ — The cube root equals exponent of $\frac{1}{3}$
17. $x^{2/3} = \sqrt[3]{x^2}$ — Denominator is root, numerator is power
18. $8^{2/3} = (\sqrt[3]{8})^2 = 2^2 = 4$
19. $27^{2/3} = (\sqrt[3]{27})^2 = 3^2 = 9$
20. $x^{1/2} \cdot x^{1/2} = x^{1/2+1/2} = x^1 = x$ — Product Rule
21. $y^{3/4} \cdot y^{1/4} = y^{3/4+1/4} = y^{4/4} = y$ — Product Rule
22. $(a^{1/4})^8 = a^{(1/4) \cdot 8} = a^2$ — Power Rule
23. $16^{-3/4} = \frac{1}{16^{3/4}} = \frac{1}{(\sqrt[4]{16})^3} = \frac{1}{2^3} = \frac{1}{8}$
24. $\sqrt{x} \cdot \sqrt[3]{x} = x^{1/2} \cdot x^{1/3} = x^{1/2+1/3} = x^{3/6+2/6} = x^{5/6}$
25. $(27x^6)^{2/3} = 27^{2/3} \cdot x^{6 \cdot 2/3} = 9 \cdot x^4 = 9x^4$